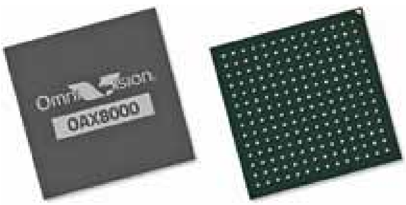
the new OPTIGA Trust Charge is a secured inductive, wireless charging solution that is usable for the Qi 1.3 wireless charging standard, which mandates strong cryptographic authentication for wireless charging devices. The IC addresses chargers for small personal electronic devices like smartphones, earbuds, tablets, wearables, or health tech devices with a charging power of up to 15W.

Infineon Technologies
www.infineon.com

Driver monitoring system ASIC with AI processing

The OAX8000, an AI-enabled, automotive application-specific integrated circuit (ASIC), is optimised for driver monitoring systems. It uses a stacked-die architecture to provide DDR3 SDRAM memory



(1Gb). The chip also integrates neural processing unit (NPU) and image signal processor that have processing speeds of up to 1.1 trillion operations per second and consume low power. The on-chip NPU is supported by TensorFlow, Caffe, MXNet, and ONNX. Additionally, this ASIC embeds quad Arm Cortex A5 CPU cores for accelerated video encoding/decoding, video analytics, and RGB/IR processing. Other applications include occupant detection and providing automotive security.

OmniVision Technologies
www.ovt.com

Laser-based sensor for connector pins' height inspection

The IX-H can take multiple measurements at once and track a target even as it moves in the scan region. The sensor's wide scan-area performs sixteen width and diameter inspections, thus carrying out jobs of multiple sensors with a single sensor. Also, the IX-H can inspect dimensions of large targets, even those

that are larger than the scan area of the sensor head. It is most suitable for performing height inspections on small targets such as connector pins.

Keyence
www.keyence.com

Dual antenna GPS/GNSS module

With dual antenna capabilities, the mosaic-H heading receiver surface-mount module delivers reliable heading and pitch, or heading and roll information, on top of centimetre-level positioning. It rapidly detects orientation angles and initialises inertial systems, which

otherwise would require movement before they can measure 3D orientation. INS initialisation with GNSS from power-up allows machine trajectory path optimisation and fully informed navigation of robotic systems immediately from mission start.

Septentrio
www.septentrio.com

Long range flash LiDAR

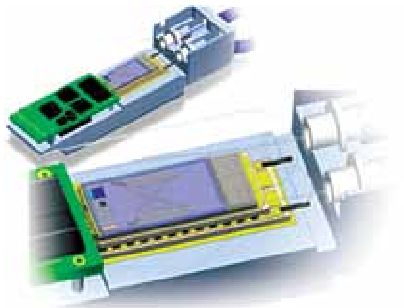
Sense Silicon, a backside-illuminated CMOS SPAD device with more than 140,000 pixels, has the ability to work seamlessly with a distributed 940nm laser array of more than 15,000 VCSELs, building together a camera-like architecture having high-resolution, eye-safe, global shutter flash LiDAR that can detect 10% reflective targets up to

200 metres away in full sunlight. This removes the need for complex motion blur correction while allowing pixel-level, frame-by-frame fusion with RGB camera data, making it suitable for automotive R&D projects involving ADAS and AV.

Sense Photonics
www.sensephotonics.com

Flip-chip DML laser for high-speed optical communication

A high-speed directly modulated laser (DML) design using flip-chip assembly technique enables a single-chip, fully integrated optical engine to be produced at wafer-scale, resulting in a low-cost, small-size 100G CWDM4 optical engine



with a small form factor. DML lasers are commonly used in 100G transceiver applications, enabling high-speed optical communication in the 2 to 10km range at speeds of 25 gigabits per second. Flip-chip assembly of electronic devices on circuit boards, MEMS, and other devices achieves advanced electrical interconnect in semiconductor architectures.

POET Technologies
www.poet-technologies.com

Intelligent 3D sensing module for smartphones

The popularity of smartphones has enabled everybody to take pictures.